

Ayush Goel

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SUMMARY

Experienced Software Engineer (5+ years) specialized in designing and scaling large distributed systems for high-availability and performance. Expertise in cloud, microservices, container orchestration (Kubernetes), Infrastructure-as-Code, and Artificial Intelligence. Proficient in modern architectural patterns (event-driven, serverless) and deeply knowledgeable about system reliability, performance tuning, and cost-optimization. Proven track record of leading complex projects, mentoring engineers, and delivering mission-critical solutions in fast-paced environments.

WORK EXPERIENCE

Software Development Engineer II

US | Apr 2025 - Present

Amazon Web Services

EC2 Team | Java, C++, Docker, Monitoring, Security, AI

- Architected and launched **Block Public Access** security feature for AWS's newer networking fleets processing **100%** of global AWS EC2 traffic, safeguarding millions of customer VPC resources from unintended internet exposure.
- Reduced code discovery time from **hours** to **≤ 5 minutes** by engineering a **RAG** system for enterprise codebase intelligence, augmenting **LLM** on AWS Bedrock using a **multi-index** retrieval strategy using OpenSearch vector embeddings, language-specific AST parsers, and MCP support to power semantic search across large distributed codebases.
- Engineered a high-performance **C++ dataplane** on AWS EC2 using **DPDK LPM6** tables to support **10M+** routing rules with a **500 MB** footprint; achieving **< 8 μs** P99 latency.

Senior Software Engineer

Salt Lake City, US | Nov 2024 - Apr 2025

Goldman Sachs

SDLC Team | Java, CI/CD, IaC, Kubernetes, AWS, Docker, Monitoring, Security

- Architected a high-throughput distributed artifact repository by redesigning the existing system for high-availability, achieving **99.99%** uptime and 10x faster dependency resolution for **100K+** daily build jobs.
- Reduced **Mean Time to Recovery** (MTTR) by **73%** for latency-sensitive trading services by architecting a **Kubernetes-native**, self-healing release framework using **Helm** and custom controllers (CRDs) for automated canary rollbacks.
- Designed a distributed pipeline lifecycle system using AWS Lambda, sharding GitLab's **1M+** daily pipelines by project for parallelized cleanup, reducing PostgreSQL storage costs by **50%** while maintaining **99.9%** API uptime during petabyte-scale deletions.
- Engineered a distributed pipeline query service bypassing GitLab API constraints with timestamp-based partitioning, achieving **15-second** latency (from **6 hours**) for **100K+** concurrent pipelines.

Software Engineering Intern

California, US | May 2023 - Dec 2023

TDK InvenSense

C++, Python, AWS IoT, ECS, Greengrass, S3, Lambda, CloudFormation

- Slashed operational costs by **30%** and tripled data ingestion rate by ingesting sensor readings with a data pipeline using AWS services and developing high-performance ROS2 C++ drivers.

Senior Software Engineer

India | Aug 2019 - Jul 2022

Unicommerce eSolutions Pvt. Ltd.

Backend Team | Early-Stage Tech Startup, E-Commerce, SaaS, Java, GraphQL, React

- Architected a distributed search infrastructure using **Elasticsearch**, designing custom **indexing pipelines** with Logstash for high-throughput ingestion, implementing **query optimization strategies** for sub-second retrieval.
- Architected high-throughput, event-driven microservices using **AWS EventBridge** for routing, **SQS** for asynchronous messaging, and global load balancers for region-level fault tolerance, supporting **30%** increase in transaction throughput.
- Designed a **sharding** approach via **consistent hashing** to handle large data sets across SQL and MongoDB, achieving a **30%** improvement in query performance and data consistency.
- Attained **99.95%** uptime with robust monitoring and alerting via **Prometheus** and **Grafana**. Enhanced system observability and automated alerts, reducing mean time to detection by **50%**.

Senior Software Engineer

India | Sep 2018 - Jun 2019

Capgemini

REST APIs, Banking, Finance, Spring, React, Microservices, Java

- Engineered a secure (**TLS**, **AES-256**), high-concurrency payment microservice for CIMB Bank, boosting transaction speed by 40% through optimized concurrent operations and minimal database lock contention.

TECHNICAL SKILLS

System Design & Architecture: Distributed Systems, Microservices Architecture, Event-Driven Design, CAP Theorem, High Availability, Fault Tolerance, CQRS, Sharding, Master-Slave Replication, Caching Strategies, Idempotent APIs, UML Diagrams

Programming: Java, Python, C++, GO, Shell Scripting, JavaScript, TypeScript

Backend Frameworks & Libraries: Spring, J2EE, Hibernate, JUnit, Mockito, GraphQL, REST, gRPC, MERN Stack

Infrastructure & DevOps: Docker, Kubernetes, EKS, Jenkins, CMake, Terraform, Grafana, Prometheus, AWS, Valgrind, gdb

Data Storage & Processing: PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch, Kafka, Database Sharding, Change Data Capture

Cloud & Serverless: AWS (EC2, ECS, EKS, S3, RDS, Lambda, SQS, VPC, CloudWatch, X-Ray), GCP, Firebase, Connection Pooling

AI/ML: SageMaker, PyTorch, scikit-learn, LangChain, pandas, MLOps (MLflow, Kubeflow), Weights&Biases, Apache Airflow, Spark

Design Patterns: Authentication (OAuth2, JWT), Rate Limiting, Retry Mechanisms, Distributed Logging, Externalized Configuration

Version Control & Development: Git, Linux, Flutter, Android, CUDA

EDUCATION

University of Pennsylvania, United States | Masters in Robotics (Concentration in Computer Science & Artificial Intelligence)

Position: Teaching Assistant for Data Structures, Algorithms and Software Design; and Operating Systems